OCI Tagging

Driving Governance, Visibility and Cost Control

Operations Advisory Team | Office of CTO
EMEA Technology Cloud Engineering

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Agenda

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Introduction

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Why Tagging Matters

Lack of organization

Resources are created without structure, making it difficult to group, filter, and manage them effectively.

Limited Visibility

It's hard to understand what resources exist, who owns them, and how they are being used.

Poor Cost Control

Without clear ownership and classification, allocating and optimizing cloud costs becomes challenging.

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OCI Tagging Overview

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OCI Tagging Overview

Tagging in OCI allows you to add key-value metadata to resources in order to organize and manage them based on business needs.

- Tags are used to organize and list resources across your cloud environment
- They are part of **OCI Identity & Access Management (IAM)**
- Tags can be applied at resource creation or later
- A tag consists of
  - ✓ Key: the tag name
  - ✓ Value: the assigned data
- They enable:
  - ✓ Resource organization
  - ✓ Search and filtering
  - ✓ Governance and automation
  - ✓ Cost tracking

https://docs.oracle.com/en-us/iaas/Content/Tagging/home.htm

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Types of Tags

Free-form Tags

- Consist of a key and a value
- **Do not belong** to a namespace
- Limited functionality
- Can be applied during resource creation or to an existing resource.
- Free-form tag keys and values are **case sensitive**
- Useful for quick, ad-hoc organization
- ***Update*** permission on the resource needed

Limitations:

- No global visibility: can’t list or track all tags centrally
- No standardization: risk of duplicates, inconsistencies, typos…
- Not supported in IAM policies
- Can't use tag variables
- Can't use predefined values
- Max 10 free-form tags per resource

Defined Tags

- Structured as namespace + key + value
- Predefined in tenancy before use by Tag Administrators
- Tag values can be **Free text** (String) or **List of values** (Enum)
- Users select from existing tag keys when applying tags
- Can be applied at creation or to existing resources
- **Keys are case insensitive, values are case sensitive**
- Support tag variables and predefined value lists
- Can enforce standardization and consistency
- Users need permissions on the resource and the tag namespace
- Tags are inherited from parent compartments

Limitations:

- Max 64 defined tags per resource
- Max 10 cost-tracking per tenancy (active and retired tags)
- Max 100 pre-defined values for a tag key per list

STANDARD TAGS

COST-TRACKING TAGS

Unlimited tags per tenancy. Total tag data size: 5 K (JSON). Includes all tag data for a single resource. Sizing is per UTF-8.

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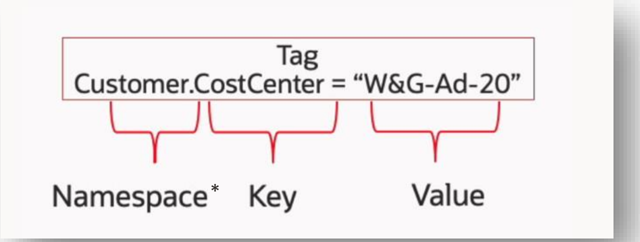
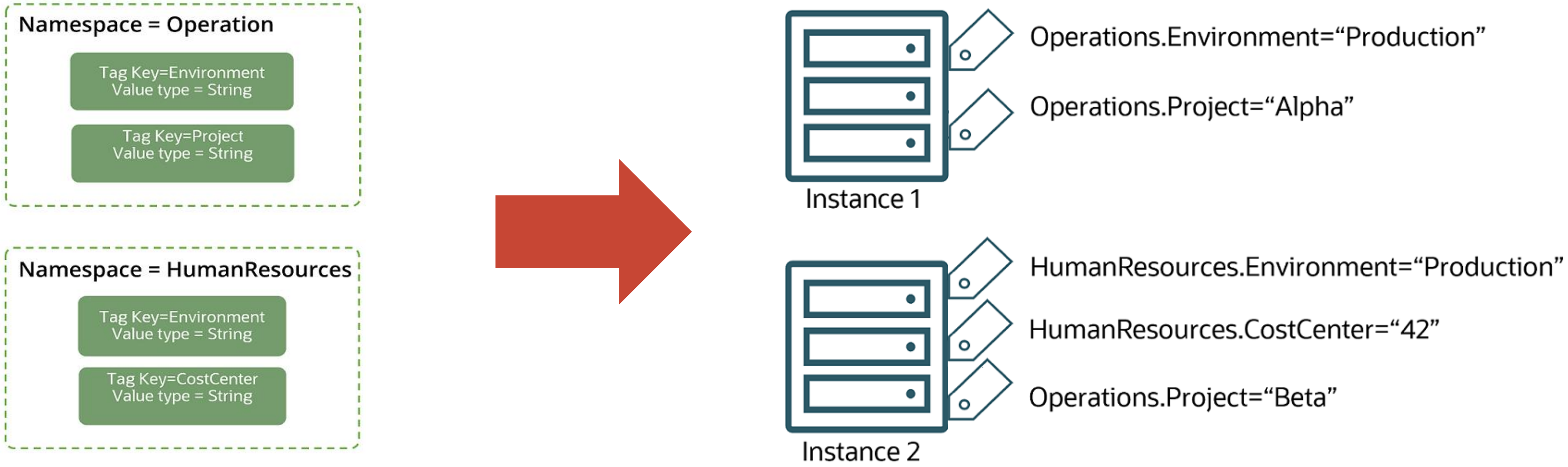
Tag Definition

By designing your tag namespaces and keys to capture critical metadata, you create a powerful framework for managing, organizing, and optimizing your cloud resources.





* Tag namespaces are not case sensitive and must be unique across the tenancy.

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Tagging Usage and Implementation

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What Can You Do with Tagging?

ACCESS CONTROL

COST MANAGEMENT

OWNERSHIP

RESOURCE GROUPING

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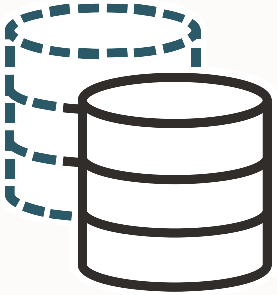
When Should Tags Be Applied?



AT RESOURCE CREATION

- Ensures consistency from day one
- Avoids missing tags
- Enables immediate cost tracking

Tagging should happen at creation time whenever possible

AFTER RESOURCE CREATION

- Improves visibility and cost allocation
- Enables cleanup and governance
- When:
  - Existing resources without tags
  - Fixing inconsistencies
  - Tagging legacy environments

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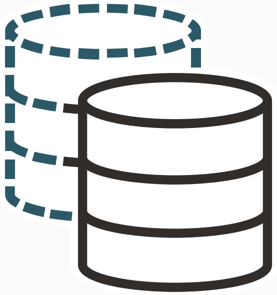
How Should Tags Be Applied?

Shift tagging to automated and enforced methods at creation time



AT RESOURCE CREATION

Manual

- OCI Console

Scripted

- OCI CLI
- SDK / API

Automated (IaC)

- Terraform / Resource Manager

Enforced

- Tag Defaults

AFTER RESOURCE CREATION

Manual

- OCI Console (edit resource)
- OCI Console (bulk edit)

Scripted

- OCI CLI
- SDK / API

Automated (IaC)

- Re-apply configuration

Only if resources are managed via IaC

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Bulk Tagging in OCI

Update tags across multiple resources at once

OCI Console*

*Menu ➔ Governance & Administration ➔ Tenancy Management ➔ Resource Explorer ➔ Select resources ➔ Actions ➔ **Manage tags***

* BulkEditTags IAM API used

Command Line Interface (OCI CLI)

Bulk edit tags on multiple resources

```
> oci iam tag bulk-edit [OPTIONS]
```

Deletes specified tag key definitions

```
> oci iam tag bulk-delete [OPTIONS]
```

Lists the resource types that support bulk tag editing

```
> oci iam tag bulk-edit-tags-resource-type list [OPTIONS]
```

CLI operations are scoped to a compartment, while the console supports cross-compartment selection

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Tagging Strategy & Governance

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Tagging Strategy Principles

- *Keep it simple, s* (**K.I.S.S.**)*
- *Start with key dimensions: Workload, Environment, Team*
- *Follow FinOps maturity: Crawl ➔ Walk ➔ Run*

What questions do you want to answer with your tags?

- Who owns this resource?
- Which team/project pays for it?
- What is it used for?

COST MANAGEMENT

- CostCenter
- Project
- Department
- Customer

➢ Enables cost allocation and FinOps practices

OPERATIONAL

- Application
- Version
- Owner
- Usage

➢ Supports day-to-day operations

GOVERNANCE

- Data classification
- Compliance (e.g. HIPAA)

➢ Ensures compliance and risk control

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Common Tagging Anti-Patterns

MISUSE OF FREE-FORM TAGS

Problem:

- No structure or governance
- Inconsistent naming

Impact:

- Poor reporting
- Not usable for FinOps

LACK OF STANDARDS

Problem:

- No defined keys or naming conventions
- Different formats across teams

Impact:

- Duplicate and inconsistent tags

NO AUTOMATION & ENFORCEMENT

Problem:

- Tags applied manually without defaults
- Relies on user discipline

Impact:

- Missing or incorrect tags
- Inconsistent data at scale

REACTIVE TAGGING

Problem:

- Tags applied after resource creation

Impact:

- Incomplete cost visibility
- Time-consuming remediation

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Tagging Best Practices for OCI

A successful tagging strategy is standardized, automated, and enforced from day one

- Standardization & Governance
  - Use Defined Tags instead of free-form tags
  - Create tag namespaces at root or high-level compartments
  - Define clear naming conventions for keys and values
- Tagging Strategy
  - Define tags based on key dimensions
  - Separate namespaces by purpose (e.g. cost, operations, governance)
- Automation & Enforcement
  - Use Tag Defaults to ensure automatic tagging
  - Implement tagging via IaC (Terraform / Resource Manager)
  - Use tag variables where applicable
- Operations & Maintenance
  - Use bulk tagging only for remediation
  - Regularly audit missing or inconsistent tags
  - Ensure tagging supports cost allocation (FinOps)

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Thank you

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